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Private Pilot for Airplane Category Airman Certification Standards

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Foreword

The U.S. Department of Transportation, Federal Aviation Administration (FAA), Office of Safety Standards, Regulatory Support Division, Airman Testing Standards Branch, has published the Private Pilot for Airplane Category Airman Certification Standards (ACS) to communicate the aeronautical knowledge, risk management, and flight proficiency standards for private pilot certification in the airplane category, single-engine land and sea; and multiengine land and sea classes.

This ACS is available for download, in PDF format, from www.faa.gov.

Comments regarding this ACS may be emailed to acsptsinquiries@faa.gov.

Material in FAA-S-ACS-6C supersedes FAA-S-ACS-6B, Private Pilot – Airplane Airman Certification Standards, Change 1.

The FAA created FAA-G-ACS-2, Airman Certification Standards Companion Guide for Pilots, to provide guidance considered relevant and useful to the community. The number of appendices in the ACS was reduced and much of the non-regulatory material was moved to the Airman Certification Standards Companion Guide for Pilots. Applicants, instructors, and evaluators should consult this companion guide to familiarize themselves with ACS procedures. FAA-G-ACS-2 is available for download, in PDF format, from www.faa.gov.

Revision History

Document #	Description	Date
FAA-S-8081-14B	Private Pilot Practical Test Standards for Airplane, (Changes 1-6)	November 2011
FAA-S-ACS-6	Private Pilot – Airplane Airman Certification Standards	June 1, 2016
FAA-S-ACS-6	Private Pilot – Airplane Airman Certification Standards (Change 1)	June 15, 2016
FAA-S-ACS-6A	Private Pilot – Airplane Airman Certification Standards (Change 1)	June 12, 2017
FAA-S-ACS-6B	Private Pilot – Airplane Airman Certification Standards	June 11, 2018
FAA-S-ACS-6B	Private Pilot – Airplane Airman Certification Standards (with Change 1)	June 6, 2019
FAA-S-ACS-6C	Private Pilot for Airplane Category Airman Certification Standards	November 2023

Major Enhancements to FAA-S-ACS-6C

- The following ACS codes have been added:

PA.I.B.K1e	PA.II.A.S4	PA.VI.D.S5	PA.VIII.E.R8
PA.I.B.K4	PA.II.B.K4	PA.VII.B.S11	PA.VIII.E.R9
PA.I.C.K2a	PA.II.B.R3	PA.VII.C.S11	PA.VIII.E.R10
PA.I.C.K2b	PA.II.B.S4	PA.VIII.A.R5	PA.VIII.E.S2
PA.I.C.K2c	PA.II.C.K4	PA.VIII.A.R6	PA.IX.A.K4
PA.I.C.K2d	PA.II.D.R4	PA.VIII.A.R7	PA.IX.A.S7
PA.I.C.K2e	PA.II.E.R5	PA.VIII.A.R8	PA.IX.A.S8
PA.I.C.K2f	PA.III.A.R4	PA.VIII.B.R5	PA.IX.C.R3
PA.I.C.K2g	PA.IV.A.R7	PA.VIII.B.R6	PA.IX.C.R4
PA.I.D.K1a	PA.IV.A.S6a	PA.VIII.B.R7	PA.IX.D.K4
PA.I.D.R7	PA.IV.A.S7a	PA.VIII.B.R8	PA.IX.D.K5
PA.I.D.S5	PA.IV.G.S6a	PA.VIII.C.R5	PA.IX.D.R2
PA.I.E.K4	PA.IV.I.S4a	PA.VIII.C.R6	PA.IX.D.R3
PA.I.H.R4	PA.IV.I.S4b	PA.VIII.C.R7	PA.IX.D.S3
PA.I.I.K1i	PA.IV.K.S6a	PA.VIII.C.R8	PA.XI.A.K6
PA.I.I.K8	PA.IV.K.S6b	PA.VIII.D.R5	PA.XI.A.K7
PA.I.I.K9	PA.IV.N.R8	PA.VIII.D.R6	PA.XI.A.K8
PA.I.I.R3	PA.IV.N.R9	PA.VIII.D.R7	PA.XI.A.R4
PA.I.I.R4	PA.IV.N.S9	PA.VIII.D.R8	PA.XI.A.R5
PA.I.I.S5	PA.VI.A.R3	PA.VIII.E.K2	PA.XI.A.R6
PA.I.I.S6	PA.VI.B.R5	PA.VIII.E.K3	PA.XI.A.R7
PA.II.A.S3	PA.VI.C.S6	PA.VIII.E.K4	

- The following ACS codes have been removed and archived. Please see the Airman Certification Standards Companion Guide for Pilots (FAA-G-ACS-2) for more information.

PA.III.A.R3	PA.VIII.E.K1b	PA.VIII.F.S3	PA.XII.A.R2
PA.IV.A.S10	PA.VIII.E.K1c	PA.IX.C.K1a	PA.XII.A.S1
PA.IV.I.S5	PA.VIII.E.K1d	PA.IX.C.K1b	PA.XII.B.R2
PA.IV.K.S7	PA.VIII.E.R2	PA.IX.C.K1c	
PA.VIII.E.K1a	PA.VIII.E.R6	PA.IX.C.K1d	

- Non-regulatory material has been moved from the appendices to the Airman Certification Standards Companion Guide for Pilots (FAA-G-ACS-2).
- Legends have been added to the Additional Ratings Task Tables.

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Introduction

Airman Certification Standards Concept

The goal of the airman certification process is to ensure the applicant possesses the knowledge, ability to manage risks, and skill consistent with the privileges of the certificate or rating being exercised, in order to act as pilot-in-command (PIC).

Safe operations in today's National Airspace System (NAS) require the integration of aeronautical knowledge, risk management, and flight proficiency standards. To accomplish these goals, the FAA drew upon the expertise of organizations and individuals across the aviation and training community to develop the ACS. The ACS integrates the elements of knowledge, risk management, and skill required for each airman certificate or rating. It thus forms a more comprehensive standard for what an applicant must know, consider, and do to demonstrate proficiency to pass the tests required for issuance of the applicable airman certificate or rating.

Area of Operation I. Preflight Preparation

Task A. Pilot Qualifications

References: 14 CFR parts 61, 68, 91; AC 68-1; FAA-H-8083-2, FAA-H-8083-3, FAA-H-8083-25

Objective: To determine the applicant exhibits satisfactory knowledge, risk management, and skills associated with airman and medical certificates including privileges, limitations, currency, and operating as pilot-in-command as a private pilot.

Knowledge: The applicant demonstrates understanding of:

- PA.I.A.K1 Certification requirements, recent flight experience, and recordkeeping.
- PA.I.A.K2 Privileges and limitations.
- PA.I.A.K3 Medical certificates: class, expiration, privileges, temporary disqualifications.
- PA.I.A.K4 Documents required to exercise private pilot privileges.
- PA.I.A.K5 Part 68 BasicMed privileges and limitations.

Risk

Management: The applicant is able to identify, assess, and mitigate risk associated with:

- PA.I.A.R1 Proficiency versus currency.
- PA.I.A.R2 Flying unfamiliar aircraft or operating with unfamiliar flight display systems and avionics.

Skills: The applicant exhibits the skill to:

- PA.I.A.S1 Apply requirements to act as pilot-in-command (PIC) under Visual Flight Rules (VFR) in a scenario given by the evaluator.

Task B. Airworthiness Requirements

References: 14 CFR parts 39, 43, 91; FAA-H-8083-2, FAA-H-8083-3, FAA-H-8083-25

Objective: To determine the applicant exhibits satisfactory knowledge, risk management, and skills associated with airworthiness requirements, including airplane certificates.

Knowledge: The applicant demonstrates understanding of:

- PA.I.B.K1 General airworthiness requirements and compliance for airplanes, including:
 - PA.I.B.K1a a. Location and expiration dates of required aircraft certificates
 - PA.I.B.K1b b. Required inspections and airplane logbook documentation
 - PA.I.B.K1c c. Airworthiness Directives and Special Airworthiness Information Bulletins
 - PA.I.B.K1d d. Purpose and procedure for obtaining a special flight permit
 - PA.I.B.K1e e. Owner/Operator and pilot-in-command responsibilities
- PA.I.B.K2 Pilot-performed preventive maintenance.
- PA.I.B.K3 Equipment requirements for day and night VFR flight, including:
 - PA.I.B.K3a a. Flying with inoperative equipment

- PA.I.B.K3b b. Using an approved Minimum Equipment List (MEL)
- PA.I.B.K3c c. Kinds of Operation Equipment List (KOEL)
- PA.I.B.K3d d. Required discrepancy records or placards
- PA.I.B.K4 Standard and special airworthiness certificates and their associated operational limitations.

Risk

Management: The applicant is able to identify, assess, and mitigate risk associated with:

- PA.I.B.R1 Inoperative equipment discovered prior to flight.

Skills:

The applicant exhibits the skill to:

- PA.I.B.S1 Locate and describe airplane airworthiness and registration information.
- PA.I.B.S2 Determine the airplane is airworthy in the scenario given by the evaluator.
- PA.I.B.S3 Apply appropriate procedures for operating with inoperative equipment in the scenario given by the evaluator.

Task C. Weather Information

References: 14 CFR part 91; AC 91-92; AIM; FAA-H-8083-2, FAA-H-8083-3, FAA-H-8083-25, FAA-H-8083-28

Objective: To determine the applicant exhibits satisfactory knowledge, risk management, and skills associated with weather information for a flight under VFR.

Note: If K2 is selected, the evaluator must assess the applicant's knowledge of at least three sub-elements.

Note: If K3 is selected, the evaluator must assess the applicant's knowledge of at least three sub-elements.

Knowledge: The applicant demonstrates understanding of:

- PA.I.C.K1 Sources of weather data (e.g., National Weather Service, Flight Service) for flight planning purposes.
- PA.I.C.K2 Acceptable weather products and resources required for preflight planning, current and forecast weather for departure, en route, and arrival phases of flight such as:
 - PA.I.C.K2a a. Airport Observations (METAR and SPECI) and Pilot Observations (PIREP)
 - PA.I.C.K2b b. Surface Analysis Chart, Ceiling and Visibility Chart (CVA)
 - PA.I.C.K2c c. Terminal Aerodrome Forecasts (TAF)
 - PA.I.C.K2d d. Graphical Forecasts for Aviation (GFA)
 - PA.I.C.K2e e. Wind and Temperature Aloft Forecast (FB)
 - PA.I.C.K2f f. Convective Outlook (AC)
 - PA.I.C.K2g g. Inflight Aviation Weather Advisories including Airmen's Meteorological Information (AIRMET), Significant Meteorological Information (SIGMET), and Convective SIGMET
- PA.I.C.K3 Meteorology applicable to the departure, en route, alternate, and destination under visual flight rules (VFR) in Visual Meteorological Conditions (VMC), including expected climate and hazardous conditions such as:
 - PA.I.C.K3a a. Atmospheric composition and stability
 - PA.I.C.K3b b. Wind (e.g., windshear, mountain wave, factors affecting wind, etc.)

PA.I.C.K3c	c. Temperature and heat exchange
PA.I.C.K3d	d. Moisture/precipitation
PA.I.C.K3e	e. Weather system formation, including air masses and fronts
PA.I.C.K3f	f. Clouds
PA.I.C.K3g	g. Turbulence
PA.I.C.K3h	h. Thunderstorms and microbursts
PA.I.C.K3i	i. Icing and freezing level information
PA.I.C.K3j	j. Fog/mist
PA.I.C.K3k	k. Frost
PA.I.C.K3l	l. Obstructions to visibility (e.g., smoke, haze, volcanic ash, etc.)
PA.I.C.K4	Flight deck instrument displays of digital weather and aeronautical information.

Risk

Management: The applicant is able to identify, assess, and mitigate risk associated with:

PA.I.C.R1	Making the go/no-go and continue/divert decisions, including:
PA.I.C.R1a	a. Circumstances that would make diversion prudent
PA.I.C.R1b	b. Personal weather minimums
PA.I.C.R1c	c. Hazardous weather conditions, including known or forecast icing or turbulence aloft
PA.I.C.R2	Use and limitations of:
PA.I.C.R2a	a. Installed onboard weather equipment
PA.I.C.R2b	b. Aviation weather reports and forecasts
PA.I.C.R2c	c. Inflight weather resources

Skills: The applicant exhibits the skill to:

PA.I.C.S1	Use available aviation weather resources to obtain an adequate weather briefing.
PA.I.C.S2	Analyze the implications of at least three of the conditions listed in K3a through K3l, using actual weather or weather conditions provided by the evaluator.
PA.I.C.S3	Correlate weather information to make a go/no-go decision.

Task D. Cross-Country Flight Planning

References: 14 CFR part 91; AIM; Chart Supplements; FAA-H-8083-2, FAA-H-8083-3, FAA-H-8083-25; NOTAMs; VFR Navigation Charts

Objective: To determine the applicant exhibits satisfactory knowledge, risk management, and skills associated with cross-country flights and VFR flight planning.

Note: Preparation, presentation, and explanation of a computer-generated flight plan is an acceptable option.

Knowledge: The applicant demonstrates understanding of:

PA.I.D.K1	Route planning, including consideration of different classes and special use airspace (SUA) and selection of appropriate and available navigation/communication systems and facilities.
PA.I.D.K1a	a. Use of an electronic flight bag (EFB), if used
PA.I.D.K2	Altitude selection accounting for terrain and obstacles, glide distance of airplane, VFR cruising altitudes, and effect of wind.
PA.I.D.K3	Calculating:
PA.I.D.K3a	a. Time, climb and descent rates, course, distance, heading, true airspeed, and groundspeed
PA.I.D.K3b	b. Estimated time of arrival, including conversion to universal coordinated time (UTC)
PA.I.D.K3c	c. Fuel requirements, including reserve
PA.I.D.K4	Elements of a VFR flight plan.
PA.I.D.K5	Procedures for filing, activating, and closing a VFR flight plan.
PA.I.D.K6	Inflight intercept procedures.

Risk

Management: The applicant is able to identify, assess, and mitigate risk associated with:

PA.I.D.R1	Pilot.
PA.I.D.R2	Aircraft.
PA.I.D.R3	Environment (e.g., weather, airports, airspace, terrain, obstacles).
PA.I.D.R4	External pressures.
PA.I.D.R5	Limitations of air traffic control (ATC) services.
PA.I.D.R6	Fuel planning.
PA.I.D.R7	Use of an electronic flight bag (EFB), if used.

Skills:

The applicant exhibits the skill to:

PA.I.D.S1	Prepare, present, and explain a cross-country flight plan assigned by the evaluator, including a risk analysis based on real-time weather, to the first fuel stop.
PA.I.D.S2	Apply pertinent information from appropriate and current aeronautical charts, Chart Supplements; Notices to Air Missions (NOTAMs) relative to airport, runway and taxiway closures; and other flight publications.
PA.I.D.S3	Create a navigation plan and simulate filing a VFR flight plan.
PA.I.D.S4	Recalculate fuel reserves based on a scenario provided by the evaluator.
PA.I.D.S5	Use an electronic flight bag (EFB), if applicable.

Task E. National Airspace System

References: 14 CFR parts 71, 91, 93; AIM; FAA-H-8083-2, FAA-H-8083-3, FAA-H-8083-25; VFR Navigation Charts

Objective: To determine the applicant exhibits satisfactory knowledge, risk management, and skills associated with National Airspace System (NAS) operations under VFR as a private pilot.

Knowledge: The applicant demonstrates understanding of:

PA.I.E.K1	Airspace classes and associated requirements and limitations.
PA.I.E.K2	Chart symbols.
PA.I.E.K3	Special use airspace (SUA), special flight rules areas (SFRA), temporary flight restrictions (TFR), and other airspace areas.
PA.I.E.K4	Special visual flight rules (VFR) requirements.

Risk

Management: The applicant is able to identify, assess, and mitigate risk associated with:

PA.I.E.R1	Various classes and types of airspace.
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Skills: The applicant exhibits the skill to:

PA.I.E.S1	Identify and comply with the requirements for basic VFR weather minimums and flying in particular classes of airspace.
PA.I.E.S2	Correctly identify airspace and operate in accordance with associated communication and equipment requirements.
PA.I.E.S3	Identify the requirements for operating in SUA or within a TFR. Identify and comply with special air traffic rules (SATR) and SFRA operations, if applicable.

Task F. Performance and Limitations

References: FAA-H-8083-1, FAA-H-8083-2, FAA-H-8083-3, FAA-H-8083-25; POH/AFM

Objective: To determine the applicant exhibits satisfactory knowledge, risk management, and skills associated with operating an airplane safely within the parameters of its performance capabilities and limitations.

Knowledge: The applicant demonstrates understanding of:

PA.I.F.K1	Elements related to performance and limitations by explaining the use of charts, tables, and data to determine performance.
PA.I.F.K2	Factors affecting performance, including:
PA.I.F.K2a	a. Atmospheric conditions
PA.I.F.K2b	b. Pilot technique
PA.I.F.K2c	c. Airplane configuration
PA.I.F.K2d	d. Airport environment
PA.I.F.K2e	e. Loading [e.g., center of gravity (CG)]
PA.I.F.K2f	f. Weight and balance
PA.I.F.K3	Aerodynamics.

Risk

Management: The applicant is able to identify, assess, and mitigate risk associated with:

PA.I.F.R1	Use of performance charts, tables, and data.
PA.I.F.R2	Airplane limitations.
PA.I.F.R3	Possible differences between calculated performance and actual performance.

Skills:	The applicant exhibits the skill to:
PA.I.F.S1	Compute the weight and balance, correct out-of-CG loading errors and determine if the weight and balance remains within limits during all phases of flight.
PA.I.F.S2	Use the appropriate airplane performance charts, tables, and data.

Task G. Operation of Systems

References: FAA-H-8083-2, FAA-H-8083-3, FAA-H-8083-23, FAA-H-8083-25; POH/AFM

Objective: To determine the applicant exhibits satisfactory knowledge, risk management, and skills associated with safe operation of systems on the airplane provided for the flight test.

Note: If K1 is selected, the evaluator must assess the applicant's knowledge of at least three sub-elements.

Knowledge:	The applicant demonstrates understanding of:
PA.I.G.K1	Airplane systems, including:
PA.I.G.K1a	a. Primary flight controls
PA.I.G.K1b	b. Secondary flight controls
PA.I.G.K1c	c. Powerplant and propeller
PA.I.G.K1d	d. Landing gear
PA.I.G.K1e	e. Fuel, oil, and hydraulic
PA.I.G.K1f	f. Electrical
PA.I.G.K1g	g. Avionics
PA.I.G.K1h	h. Pitot-static, vacuum/pressure, and associated flight instruments
PA.I.G.K1i	i. Environmental
PA.I.G.K1j	j. Deicing and anti-icing
PA.I.G.K1k	k. Water rudders (ASES, AMES)
PA.I.G.K1l	l. Oxygen system
PA.I.G.K2	Indications of and procedures for managing system abnormalities or failures.

Risk

Management: The applicant is able to identify, assess, and mitigate risk associated with:

PA.I.G.R1	Detection of system malfunctions or failures.
PA.I.G.R2	Management of a system failure.
PA.I.G.R3	Monitoring and management of automated systems.

Skills:	The applicant exhibits the skill to:
PA.I.G.S1	Operate at least three of the systems listed in K1a through K1l appropriately.
PA.I.G.S2	Complete the appropriate checklist(s).

Task H. Human Factors

References: AIM; FAA-H-8083-2, FAA-H-8083-3, FAA-H-8083-25

Objective: To determine the applicant exhibits satisfactory knowledge, risk management, and skills associated with personal health, flight physiology, and aeromedical and human factors related to safety of flight.

Knowledge: The applicant demonstrates understanding of:

- | | |
|-------------------|---|
| <i>PA.I.H.K1</i> | Symptoms, recognition, causes, effects, and corrective actions associated with aeromedical and physiological issues, including: |
| <i>PA.I.H.K1a</i> | a. Hypoxia |
| <i>PA.I.H.K1b</i> | b. Hyperventilation |
| <i>PA.I.H.K1c</i> | c. Middle ear and sinus problems |
| <i>PA.I.H.K1d</i> | d. Spatial disorientation |
| <i>PA.I.H.K1e</i> | e. Motion sickness |
| <i>PA.I.H.K1f</i> | f. Carbon monoxide poisoning |
| <i>PA.I.H.K1g</i> | g. Stress |
| <i>PA.I.H.K1h</i> | h. Fatigue |
| <i>PA.I.H.K1i</i> | i. Dehydration and nutrition |
| <i>PA.I.H.K1j</i> | j. Hypothermia |
| <i>PA.I.H.K1k</i> | k. Optical illusions |
| <i>PA.I.H.K1l</i> | l. Dissolved nitrogen in the bloodstream after scuba dives |
| <i>PA.I.H.K2</i> | Regulations regarding use of alcohol and drugs. |
| <i>PA.I.H.K3</i> | Effects of alcohol, drugs, and over-the-counter medications. |
| <i>PA.I.H.K4</i> | Aeronautical Decision-Making (ADM) to include using Crew Resource Management (CRM) or Single-Pilot Resource Management (SRM), as appropriate. |

Risk

Management: The applicant is able to identify, assess, and mitigate risk associated with:

- | | |
|------------------|--|
| <i>PA.I.H.R1</i> | Aeromedical and physiological issues. |
| <i>PA.I.H.R2</i> | Hazardous attitudes. |
| <i>PA.I.H.R3</i> | Distractions, task prioritization, loss of situational awareness, or disorientation. |
| <i>PA.I.H.R4</i> | Confirmation and expectation bias. |

Skills: The applicant exhibits the skill to:

- | | |
|------------------|---|
| <i>PA.I.H.S1</i> | Associate the symptoms and effects for at least three of the conditions listed in K1a through K1l with the cause(s) and corrective action(s). |
| <i>PA.I.H.S2</i> | Perform self-assessment, including fitness for flight and personal minimums, for actual flight or a scenario given by the evaluator. |

Task I. Water and Seaplane Characteristics, Seaplane Bases, Maritime Rules, and Aids to Marine Navigation (ASES, AMES)

References: AIM; Chart Supplements; FAA-H-8083-2, FAA-H-8083-3, FAA-H-8083-23, FAA-H-8083-25; POH/AFM; USCG Navigation Rules

Objective: To determine the applicant exhibits satisfactory knowledge, risk management, and skills associated with water and seaplane characteristics, seaplane bases, maritime rules, and aids to marine navigation.

Knowledge: The applicant demonstrates understanding of:

- PA.I.I.K1 The characteristics of a water surface as affected by features, such as:
 - PA.I.I.K1a a. Size and location
 - PA.I.I.K1b b. Protected and unprotected areas
 - PA.I.I.K1c c. Surface wind
 - PA.I.I.K1d d. Direction and strength of water current
 - PA.I.I.K1e e. Floating and partially submerged debris
 - PA.I.I.K1f f. Sandbars, islands, and shoals
 - PA.I.I.K1g g. Vessel traffic and wakes
 - PA.I.I.K1h h. Other characteristics specific to the area
 - PA.I.I.K1i i. Direction and height of waves
- PA.I.I.K2 Float and hull construction, and its effect on seaplane performance.
- PA.I.I.K3 Causes of porpoising and skipping, and the pilot action needed to prevent or correct these occurrences.
- PA.I.I.K4 How to locate and identify seaplane bases on charts or in directories.
- PA.I.I.K5 Operating restrictions at various bases.
- PA.I.I.K6 Right-of-way, steering, and sailing rules pertinent to seaplane operation.
- PA.I.I.K7 Marine navigation aids, such as buoys, beacons, lights, sound signals, and range markers.
- PA.I.I.K8 Naval vessel protection zones.
- PA.I.I.K9 No wake zones.

Risk

Management: The applicant is able to identify, assess, and mitigate risk associated with:

- PA.I.I.R1 Local conditions.
- PA.I.I.R2 Impact of marine traffic.
- PA.I.I.R3 Right-of-way and sailing rules pertinent to seaplane operations.
- PA.I.I.R4 Limited services and assistance available at seaplane bases.

Skills: The applicant exhibits the skill to:

- PA.I.I.S1 Assess the water surface characteristics for the proposed flight.
- PA.I.I.S2 Identify restrictions at local seaplane bases.

Area of Operation I. Preflight Preparation

- PA.I.I.S3* Identify marine navigation aids.
- PA.I.I.S4* Describe correct right-of-way, steering, and sailing operations.
- PA.I.I.S5* Explain how float and hull construction can affect seaplane performance.
- PA.I.I.S6* Describe how to correct for porpoising and skipping.

Area of Operation XI. Night Operations

Task A. Night Operations

References: AIM; FAA-H-8083-2, FAA-H-8083-3, FAA-H-8083-25; POH/AFM

Objective: To determine the applicant exhibits satisfactory knowledge and risk management associated with night operations.

Note: For applicants that reside in Alaska, refer to 14 CFR part 61, section 61.110.

Knowledge: The applicant demonstrates understanding of:

- PA.XI.A.K1 Physiological aspects of vision related to night flying.
- PA.XI.A.K2 Lighting systems identifying airports, runways, taxiways and obstructions, as well as pilot controlled lighting.
- PA.XI.A.K3 Airplane equipment and lighting requirements for night operations.
- PA.XI.A.K4 Personal equipment essential for night flight.
- PA.XI.A.K5 Night orientation, navigation, chart reading techniques and methods for maintaining night vision effectiveness.
- PA.XI.A.K6 Night taxi operations.
- PA.XI.A.K7 Interpretation of traffic position and direction based solely on position lights.
- PA.XI.A.K8 Visual illusions at night.

Risk

Management: The applicant is able to identify, assess, and mitigate risk associated with:

- PA.XI.A.R1 Collision hazards.
- PA.XI.A.R2 Distractions, task prioritization, loss of situational awareness, or disorientation.
- PA.XI.A.R3 Effect of visual illusions and night adaptation during all phases of night flying.
- PA.XI.A.R4 Runway incursion.
- PA.XI.A.R5 Night currency versus proficiency.
- PA.XI.A.R6 Weather considerations specific to night operations.
- PA.XI.A.R7 Inoperative equipment.

Skills: The applicant exhibits the skill to:
[Intentionally left blank].

Area of Operation XII. Postflight Procedures

Task A. After Landing, Parking, and Securing (ASEL, AMEL)

References: FAA-H-8083-2, FAA-H-8083-3, FAA-H-8083-25; POH/AFM

Objective: To determine the applicant exhibits satisfactory knowledge, risk management, and skills associated with after landing, parking, and securing procedures.

Knowledge: The applicant demonstrates understanding of:

PA.XII.A.K1 Airplane shutdown, securing, and postflight inspection.

PA.XII.A.K2 Documenting in-flight/postflight discrepancies.

Risk

Management: The applicant is able to identify, assess, and mitigate risk associated with:

PA.XII.A.R1 Activities and distractions.

PA.XII.A.R2 [Archived]

PA.XII.A.R3 Airport specific security procedures.

PA.XII.A.R4 Disembarking passengers safely on the ramp and monitoring passenger movement while on the ramp.

Skills: The applicant exhibits the skill to:

PA.XII.A.S1 [Archived]

PA.XII.A.S2 Park in an appropriate area, considering the safety of nearby persons and property.

PA.XII.A.S3 Complete the appropriate checklist(s).

PA.XII.A.S4 Conduct a postflight inspection and document discrepancies and servicing requirements, if any.

PA.XII.A.S5 Secure the airplane.

Task B. Seaplane Post-Landing Procedures (ASES, AMES)

References: FAA-H-8083-2, FAA-H-8083-3, FAA-H-8083-23, FAA-H-8083-25; POH/AFM

Objective: To determine the applicant exhibits satisfactory knowledge, risk management, and skills associated with anchoring, docking, mooring, and ramping/beaching.

Note: The evaluator must select at least one after-landing procedure (anchoring, docking and mooring, or ramping/beaching).

Knowledge: The applicant demonstrates understanding of:

PA.XII.B.K1 Mooring.

PA.XII.B.K2 Docking.

PA.XII.B.K3 Anchoring.

PA.XII.B.K4 Beaching/ramping.

PA.XII.B.K5 Postflight inspection, recording of in-flight/postflight discrepancies.

Risk

Management: The applicant is able to identify, assess, and mitigate risk associated with:

- PA.XII.B.R1* Activities and distractions.
- PA.XII.B.R2* [Archived]
- PA.XII.B.R3* Seaplane base specific security procedures, if applicable.
- PA.XII.B.R4* Disembarking passengers safely on the ramp and monitoring passenger movement while on the ramp.

Skills: The applicant exhibits the skill to:

- PA.XII.B.S1* If anchoring, select a suitable area considering seaplane movement, water depth, tide, wind, and weather changes. Use an adequate number of anchors and lines of sufficient strength and length to ensure the seaplane's security.
- PA.XII.B.S2* If not anchoring, approach the dock/mooring buoy or beach/ramp in the proper direction and at a safe speed, considering water depth, tide, current, and wind.
- PA.XII.B.S3* Complete the appropriate checklist(s).
- PA.XII.B.S4* Conduct a postflight inspection and document discrepancies and servicing requirements, if any.
- PA.XII.B.S5* Secure the seaplane considering the effect of wind, waves, and changes in water level, or comply with applicable after landing, parking, and securing procedures if operating an amphibious airplane on land.

Appendix 1: Practical Test Roles, Responsibilities, and Outcomes

Eligibility Requirements for a Private Pilot Certificate

The prerequisite requirements and general eligibility for a practical test and the specific requirements for the original issuance of a Private Pilot Certificate in the airplane category can be found in 14 CFR part 61, sections 61.39(a) and 61.103.

In accordance with 14 CFR part 61, section 61.39, the applicant must pass the airman knowledge test before taking the practical test, if applicable to the certificate or rating sought.

Note: *An applicant seeking to add an additional category or class to an existing certificate must comply with 14 CFR section 61.63, as applicable.*

Private Pilot Airplane Knowledge Test Table

For an initial Private Pilot Certificate, applicants must pass the knowledge test listed in the table below as a prerequisite for the practical test.

Test Code	Test Name	Number of Questions	Age	Allotted Time	Passing Score
PAR	Private Pilot Airplane	60	15	2.5	70

Use of the ACS During a Practical Test

The practical test is conducted in accordance with the ACS and FAA regulations that are current as of the date of the test.

The Areas of Operation in this ACS align with the Areas of Operation found in 14 CFR part 61, section 61.107(b). Each Area of Operation includes Tasks appropriate to that Area of Operation. Each Task contains an Objective stating what the applicant must know, consider, and/or do. The ACS then lists the aeronautical knowledge, risk management, and skill elements relevant to the specific Task, along with the conditions and standards for acceptable performance. The ACS uses Notes to emphasize special considerations.

During the ground and flight portion of the practical test, the FAA expects evaluators to assess the applicant's mastery of the topic in accordance with the level of learning most appropriate for the specified Task. The oral questioning will continue throughout the entire practical test. For some topics, the evaluator will ask the applicant to describe or explain. For other items, the evaluator will assess the applicant's understanding by providing a scenario that requires the applicant to appropriately apply and/or correlate knowledge, experience, and information to the circumstances of the given scenario. The flight portion of the practical test requires the applicant to demonstrate knowledge, risk management, flight proficiency, and operational skill in accordance with the ACS.

The elements within each Task in this ACS are coded according to a scheme that includes four components. For example, PA.I.C.K2:

PA = Applicable ACS

I = Area of Operation

C = Task

K2 = Task element (in this example, Knowledge 2)

There is no requirement for an evaluator to test every knowledge and risk management element in a Task; rather the evaluator has discretion to sample as needed to ensure the applicant's mastery of that Task. The required minimum elements to be tested from each applicable Task include:

- any elements in which the applicant was shown to be deficient on the knowledge test, as applicable;
- at least one knowledge element;
- at least one risk management element; and
- all skill elements unless otherwise noted.

Note: *The Airman Knowledge Test Report (AKTR) lists ACS codes that correlate to a specific Task element for a given Area of Operation for any incorrect responses on the knowledge test.*

Knowledge and risk management elements are primarily evaluated during the knowledge testing phase of the airman certification process. The evaluator administering the practical test has the discretion to combine Tasks/elements as appropriate to testing scenarios.

Unless otherwise noted in the Task, the evaluator must test each item in the skills section by observing the applicant perform each one. As safety of flight conditions permit, the evaluator should use questions during flight to test knowledge and risk management elements not evident in the demonstrated skills. To the greatest extent practicable, evaluators should test the applicant's ability to apply and correlate information and use rote questions only when they are appropriate for the material being tested.

If the Task includes a knowledge or risk element with sub-elements, the evaluator may choose the primary element and select at least one sub-element to satisfy the requirement. Selection of the sub-element satisfies the requirement for one element unless otherwise noted.

For example, an evaluator who chooses PA.I.H.K1 may select a sub-element such as PA.I.H.K1e to satisfy the requirement to select one knowledge element.

The References for each Task indicate the source material for Task elements. For example, in the Task element "Acceptable weather products and resources required for preflight planning, current and forecast weather for departure, en route, and arrival phases of flight such as:" (PA.I.C.K2), the applicant should be prepared for questions on any weather product presented in the references for that Task.

The FAA encourages applicants and instructors to use the ACS when preparing for the airman knowledge tests and practical tests. Evaluators must conduct the practical test in accordance with the current ACS and FAA regulations pursuant to 14 CFR part 61, section 61.43. If an applicant is entitled to credit for Areas of Operation previously passed as indicated on a Notice of Disapproval of Application or Letter of Discontinuance, evaluators shall use the ACS currently in effect on the date of the test.

The ground portion of the practical test allows the evaluator to determine whether the applicant is sufficiently prepared to advance to the flight portion of the practical test. The applicant must pass the ground portion of the practical test before beginning the flight portion. The oral questioning will continue throughout the entire practical test.

Combined Private/Instrument Test

Applicants for a combined Private Pilot Certificate with Instrument Rating, in accordance with 14 CFR part 61, section 61.65(a) and (g), must pass all areas designated in the Private Pilot for Airplane ACS and the Instrument Rating – Airplane ACS. Evaluators need not duplicate Tasks. For example, only one preflight demonstration would be required; however, the Preflight Task from the Instrument Rating – Airplane ACS would be more extensive than the Preflight Task from the Private Pilot for Airplane ACS to ensure readiness for Instrument Flight Rules (IFR) flight. Applicants for a combined test must present the applicable test reports.

A combined certificate and rating evaluation should be treated as one practical test, requiring only one application and resulting in only one temporary certificate, disapproval notice, or letter of discontinuance, as applicable. Failure of any Task will result in a failure of the entire test and application. Therefore, even if the deficient maneuver was instrument related and the performance of all visual flight rules (VFR) Tasks was determined to be satisfactory, the applicant will receive a notice of disapproval.

Instructor Responsibilities

The instructor trains and qualifies the applicant to meet the established standards for knowledge, risk management, and skill elements in all Tasks appropriate to the certificate and rating sought. The instructor should use this ACS and its references when preparing the applicant to take the practical test and when retraining the applicant to proficiency in any subject(s) missed on the knowledge test.

Evaluator Responsibilities

An evaluator includes the following:

- Aviation Safety Inspector (ASI);
- Pilot examiner (other than administrative pilot examiners);
- Training center evaluator (TCE);
- Chief instructor, assistant chief instructor, or check instructor of pilot school holding examining authority; or

- Instrument Flight Instructor (CFII) conducting an instrument proficiency check (IPC).

The evaluator who conducts the practical test verifies the applicant has met the aeronautical experience requirements specified for a certificate or rating before administering the test. During the practical test, the evaluator determines whether the applicant meets the established standards of aeronautical knowledge, risk management, and skills for the Tasks in the appropriate ACS.

The evaluator must develop a plan of action (POA) that includes all required Areas of Operation and Tasks and administer each practical test in English. The POA must include scenario(s) that evaluate as many of the required Areas of Operation and Tasks as possible. As a scenario unfolds during the test, the evaluator will introduce problems and simulate emergencies that test the applicant's ability. The evaluator has the discretion to modify the POA to accommodate unexpected situations as they arise or suspend and later resume a scenario to assess certain Tasks.

Prior to and throughout the evaluation, the evaluator ensures the applicant meets the FAA Aviation English Language Standard (AELS). An applicant must be able to communicate in English in a discernible and understandable manner with air traffic control (ATC), pilots, and others involved in preparing an aircraft for flight and operating an aircraft in flight. This communication may or may not involve radio communications. An applicant for an FAA certificate or rating issued in accordance with 14 CFR parts 61, 63, 65, or 107 who cannot hear or speak due to a medical deficiency may be eligible for an FAA certificate with specific operational limitations.

If the applicant's ability to meet the FAA AELS comes into question before starting the practical test, the evaluator will not begin the practical test. An evaluator other than an ASI will check the box, "Referred to FSO for Aviation English Language Standard Determination," located on the bottom of page 2 of the applicant's FAA Form 8710-1, Airman Certificate and/or Rating Application, or FAA Form 8710-11, Airman Certificate and/or Rating Application - Sport Pilot, as applicable. The evaluator will refer the applicant to the appropriate Flight Standards Office (FSO).

If the applicant's ability to meet the FAA AELS comes into question after the practical test begins, an evaluator who other than an ASI will discontinue the practical test and check the box, "Referred to FSO for Aviation English Language Standard Determination," on the application. The evaluator will also issue FAA Form 8060-5, Notice of Disapproval of Application, with the comment "Does Not Demonstrate FAA AELS" in addition to any unsatisfactory Task(s). The evaluator will refer the applicant to the appropriate FSO. ASIs conducting the practical test may assess an applicant's English language proficiency in accordance with FAA Order 8900.1.

In either case, the evaluator must complete and submit the application file through normal application procedures and evaluators other than an ASI notify the appropriate FSO of the referral.

If the ability of an FAA certificated airman comes into question prior to or during a required regulatory check (e.g., proficiency check) the evaluator other than an ASI will not continue the check or provide an endorsement indicating completion. The evaluator will refer the airman to the jurisdictional FAA field office for further determination of ability to meet the FAA AELS.

For additional information, reference AC 60-28, FAA English Language Standard for an FAA Certificate issued under 14 CFR parts 61, 63, 65, and 107, as amended.

The abbreviation(s) within parentheses immediately following a Task refers to the category and/or class airplane appropriate to that Task. The absence of a class indicates the Task is for all classes. The meaning of each abbreviation is as follows:

- ASEL—Airplane – Single-Engine Land
- ASES—Airplane – Single-Engine Sea
- AMEL—Airplane – Multiengine Land
- AMES—Airplane – Multiengine Sea

Note: *Note: When administering a test, the Tasks appropriate to the class airplane (ASEL, ASES, AMEL, or AMES) used for the test must be included in the plan of action.*

Possible Outcomes of the Test

A practical test has three possible outcomes: (1) Temporary Airman Certificate (satisfactory), (2) Notice of Disapproval of Application (unsatisfactory), or (3) Letter of Discontinuance.

If the evaluator determines that a Task is incomplete, or the outcome is uncertain, the evaluator must require the applicant to repeat that Task, or portions of that Task. This provision does not mean that instruction, practice, or the repetition of an unsatisfactory Task is permitted during the practical test.

Satisfactory Performance

Refer to 14 CFR part 61, section 61.43, for satisfactory performance requirements.

Satisfactory performance will result in the issuance of a temporary certificate.

Unsatisfactory Performance

If, in the judgment of the evaluator, the applicant does not meet the standards for any Task, the applicant fails the Task and associated Area of Operation and the evaluator issues a Notice of Disapproval of Application. The evaluator lists the Area(s) of Operation in which the applicant did not meet the standard, any Area(s) of Operation not tested, and the number of practical test failures. The evaluator should also list the Tasks failed or Tasks not tested within any unsatisfactory or partially completed Area(s) of Operation. 14 CFR part 61, section 61.43(c)-(f) provides additional unsatisfactory performance requirements and parameters.

Typical areas of unsatisfactory performance and grounds for disqualification include:

- Any action or lack of action by the applicant that requires corrective intervention by the evaluator to maintain safe flight.
- Failure to use proper and effective visual scanning techniques to clear the area before and while performing maneuvers.
- Consistently exceeding tolerances stated in the skill elements of the Task.
- Failure to take prompt corrective action when tolerances are exceeded.
- Failure to exercise risk management.

The evaluator or the applicant may end the test if the applicant fails a Task. The evaluator may continue the test only with the consent of the applicant. The applicant receives credit only for those Areas of Operation and the associated Tasks performed satisfactorily.

Letter of Discontinuance

Refer to 14 CFR part 61, section 61.43(e)(2) for conditions to issue a letter of discontinuance.

If discontinuing a practical test for reasons other than unsatisfactory performance (e.g., equipment failure, weather, illness), the evaluator must return all test paperwork to the applicant. The evaluator must prepare, sign, and issue a Letter of Discontinuance that lists those Areas of Operation the applicant successfully completed and the time period remaining to complete the test to receive credit for previously completed Areas of Operation. The evaluator should advise the applicant to present the Letter of Discontinuance to the evaluator when the practical test resumes in order to receive credit for the items successfully completed. The Letter of Discontinuance becomes part of the applicant's certification file.

Time Limit and Credit after a Discontinued Practical Test

Refer to 14 CFR part 61, sections 61.39(f) and 61.43(f) after issuance of a Letter of Discontinuance or Notice of Disapproval of Application.

Additional Rating Task Table

For an applicant who holds at least a Private Pilot Certificate and seeks an additional airplane category and/or class rating at the private pilot level, the evaluator must evaluate that applicant in the Areas of Operation and Tasks listed in the Additional Rating Task Table. The evaluator may evaluate the applicant's competence in the remaining Areas of Operation and Tasks.

If the applicant holds two or more category or class ratings at the private level, and the ratings table indicates different Task requirements, the "least restrictive" entry applies. For example, if "All" and "None" are indicated for one Area of Operation, the "None" entry applies. If the table indicates "B" and "B, C" the "B" entry applies.

Addition of an Airplane Single-Engine Land Rating to an Existing Private Pilot Certificate

The table below indicates the required Tasks for each Area of Operation tested in accordance with this ACS.

Legend	
ASES	Airplane – Single-Engine Sea
AMEL	Airplane – Multiengine Land
AMES	Airplane – Multiengine Sea
RH	Rotorcraft – Helicopter
RG	Rotorcraft – Gyroplane
PL	Powered-Lift

Area of Operation	Private Pilot Rating(s) Held								
	ASES	AMEL	AMES	RH	RG	Glider	Balloon	Airship	PL
I	F,G	F,G	F,G	F,G	F,G	D,F,G	D,F,G	F,G	F,G
II	A,B,D,F	A,B,F	A,B,D,F	A,B,C,D, F	A,B,C,D, F	A,B,C,D, F	A,B,C,D, F	A,B,C,D, F	A,B,C,D, F
III	B	None	B	B	B	B	B	B	B
IV	A,B,C, D,E,F	A,B,C, D,E,F,M	A,B,C,D, E,F,M	A,B,C,D, E,F,M,N	A,B,C,D, E,F,M,N	A,B,C,D, E,F,M,N	A,B,C,D, E,F,M,N	A,B,C,D, E,F,M,N	A,B,C,D, E,F,M,N
V	None	None	None	All	A	All	All	All	All
VI	None	None	None	None	None	A,B,C,D	A,B,C,D	None	None
VII	None	None	None	All	All	All	All	All	All
VIII	None	None	None	All	All	All	All	All	All
IX	A,B,C	A,B,C	A,B,C	A,B,C,D	A,B,C,D	A,B,C,D	A,B,C,D	A,B,C,D	A,B,C,D
X	None	None	None	None	None	None	None	None	None
XI	None	None	None	None	None	A	A	A	A
XII	A	None	A	A	A	A	A	A	A

Addition of an Airplane Single-Engine Sea Rating to an Existing Private Pilot Certificate

The table below indicates the required Tasks for each Area of Operation tested in accordance with this ACS.

Legend	
ASEL	Airplane – Single-Engine Land
AMEL	Airplane – Multiengine Land
AMES	Airplane – Multiengine Sea
RH	Rotorcraft – Helicopter
RG	Rotorcraft – Gyroplane
PL	Powered-Lift

Area of Operation	Private Pilot Rating(s) Held								
	ASEL	AMEL	AMES	RH	RG	Glider	Balloon	Airship	PL
I	F,G,I	F,G,I	F,G	F,G,I	D,F,G,I	D,F,G,I	D,F,G,I	D,F,G,I	F,G,I
II	A,B,E,F	A,B,E,F	A,B,F	A,B,C,E,F	A,B,C,E,F	A,B,C,E,F	A,B,C,E,F	A,B,C,E,F	A,B,C,E,F
III	B	B	None	B	B	B	B	B	B
IV	A,B,G,H,I,J,K,L	A,B,G,H,I,J,K,L,M	A,B,M	A,B,G,H,I,J,K,L,M,N	A,B,G,H,I,J,K,L,M,N	A,B,G,H,I,J,K,L,M,N	A,B,G,H,I,J,K,L,M,N	A,B,G,H,I,J,K,L,M,N	A,B,G,H,I,J,K,L,M,N
V	None	None	None	All	A	All	All	All	A
VI	None	None	None	None	None	All	All	None	None
VII	None	None	None	All	All	All	All	All	All
VIII	None	None	None	All	All	All	All	All	All
IX	A,B	A,B	A,B	A,B,C,D	A,B,C,D	A,B,C,D	A,B,C,D	A,B,C,D	A,B,C,D
X	None	None	None	None	None	None	None	None	None
XI	None	None	None	None	None	All	All	All	None
XII	B	B	None	B	B	B	B	B	B

Addition of an Airplane Multiengine Land Rating to an Existing Private Pilot Certificate

The table below indicates the required Tasks for each Area of Operation tested in accordance with this ACS.

Legend	
ASEL	Airplane – Single-Engine Land
ASES	Airplane – Single-Engine Sea
AMES	Airplane – Multiengine Sea
RH	Rotorcraft – Helicopter
RG	Rotorcraft – Gyroplane
PL	Powered-Lift

Area of Operation	Private Pilot Rating(s) Held								
	ASEL	ASES	AMES	RH	RG	Glider	Balloon	Airship	PL
I	F,G	F,G	F,G	F,G	F,G	D,F,G	D,F,G	F,G	F,G
II	A,B,C,D,F	A,B,C,D,F	A,D	A,B,C,D,F	A,B,C,D,F	A,B,C,D,F	A,B,C,D,F	A,B,C,D,F	A,B,C,D,F
III	None	B	B	B	B	B	B	B	B
IV	A,B,E,F	A,B,E,F	A,B,E,F	A,B,E,F,N	A,B,E,F,N	A,B,E,F,N	A,B,E,F,N	A,B,E,F,N	A,B,E,F,N
V	A	A	None	All	A	All	All	All	A
VI	None	None	None	None	None	All	All	None	None
VII	All	All	None	All	All	All	All	All	All
VIII	None	None	None	All	All	All	All	All	All
IX	E,F,G	E,F,G	None	A,C,E,F,G	A,C,E,F,G	A,C,E,F,G	A,C,E,F,G	A,C,E,F,G	A,C,E,F,G
X*	All	All	None	All	All	All	All	All	All
XI	None	None	None	None	None	All	All	All	None
XII	None	A	A	A	A	A	A	A	A

*Tasks C and D requirements do not apply to applicants who do not hold an instrument rating or who have previously demonstrated instrument proficiency in a multiengine airplane.

Addition of an Airplane Multiengine Sea Rating to an existing Private Pilot Certificate

The table below indicates the required Tasks for each Area of Operation tested in accordance with this ACS.

Legend	
ASEL	Airplane – Single-Engine Land
ASES	Airplane – Single-Engine Sea
AMEL	Airplane – Multiengine Land
RH	Rotorcraft – Helicopter
RG	Rotorcraft – Gyroplane
PL	Powered-Lift

Area of Operation	Private Pilot Rating(s) Held								
	ASEL	ASES	AMEL	RH	RG	Glider	Balloon	Airship	PL
I	F,G,I	F,G	F,G,I	F,G,I	F,G,I	D,F,G,I	D,F,G,I	F,G,I	F,G,I
II	A,B,E,F	A,B,E,F	A,B,E,F	A,B,C,E,F	A,B,C,E,F	A,B,C,E,F	A,B,C,E,F	A,B,C,E,F	A,B,C,E,F
III	None	None	None	B	B	B	B	B	B
IV	A,B,G,H,I,J,K,L	A,B	A,B,G,H,I,J,K,L	A,B,G,H,I,J,K,L,N	A,B,G,H,I,J,K,L,N	A,B,G,H,I,J,K,L,N	A,B,G,H,I,J,K,L,N	A,B,G,H,I,J,K,L,N	A,B,G,H,I,J,K,L,N
V	A	A	None	All	A	All	All	All	A
VI	None	None	None	None	None	All	All	None	None
VII	All	All	None	All	All	All	All	All	All
VIII	None	None	None	All	All	All	All	All	All
IX	E,F,G	E,F,G	None	A,C,E,F,G	A,C,E,F,G	A,C,E,F,G	A,C,E,F,G	A,C,E,F,G	A,C,E,F,G
X*	All	All	None	All	All	All	All	All	All
XI	None	None	None	None	None	All	All	All	None
XII	B	None	B	B	B	B	B	B	B

*Tasks C and D requirements do not apply to applicants who do not hold an instrument rating or who have previously demonstrated instrument proficiency in a multiengine airplane.

Note: An applicant who holds a private airplane multiengine land rating without a center thrust limitation need not supply a seaplane with propeller feathering capability when testing to add a private airplane multiengine sea rating.

Removal of the “Airplane Multiengine VFR Only” Limitation

The removal of the “Airplane Multiengine VFR Only” limitation, at the private pilot certificate level, requires an applicant to satisfactorily perform the following Area of Operation and Tasks from the Private Pilot for Airplane Category ACS in a multiengine airplane that has a manufacturer’s published minimum control speed (V_{MC}).

X. Multiengine Operations

Task C: Engine Failure During Flight (by Reference to Instruments) (AMEL, AMES)

Task D: Instrument Approach and Landing with an Inoperative Engine (Simulated) (by Reference to Instruments) (AMEL, AMES)

Removal of the “Limited to Center Thrust” Limitation

Applicants who complete the practical test for the AMEL rating in an aircraft that does not have a manufacturer’s published V_{MC} receive a certificate with a “Limited to Center Thrust” limitation. When conducting a practical test for the purpose of removing the “Limited to Center Thrust” limitation, the evaluator must test the applicant on the multiengine Tasks identified in the table below in a multiengine airplane that has a manufacturer’s published V_{MC} speed. This speed appears on the type certificate data sheet (TCDS) or in the Airplane Flight Manual (AFM). Removal of the limitation under 14 CFR parts 121, 135, or 142 occurs in accordance with an approved curriculum or training program. An applicant who holds an airplane instrument rating and has not demonstrated instrument proficiency in a multiengine airplane with a published V_{MC} shall complete the additional Tasks listed under Removal of the “Airplane Multiengine VFR Only” Limitation section.

IX. Emergency Operations

Task E: Engine Failure During Takeoff Before V_{MC} (Simulated) (AMEL, AMES)

Task F: Engine Failure After Liftoff (Simulated) (AMEL, AMES)

Task G: Approach and Landing with an Inoperative Engine (Simulated) (AMEL, AMES)

X. Multiengine Operations

Task A: Maneuvering with One Engine Inoperative (AMEL, AMES)

Task B: V_{MC} Demonstration (AMEL, AMES)

Appendix 2: Safety of Flight

General

Safety of flight must be the prime consideration at all times. The evaluator, applicant, and crew must be continually alert for other traffic. If performing aspects of a given maneuver, such as emergency procedures, would jeopardize safety, the evaluator will ask the applicant to simulate that portion of the maneuver. The evaluator will assess the applicant's use of visual scanning and collision avoidance procedures throughout the entire test.

Stall and Spin Awareness

During flight training and testing, the applicant and the instructor or evaluator must always recognize and avoid operations that could lead to an inadvertent stall or spin and inadvertent loss of control.

Use of Checklists

Throughout the practical test, the applicant is evaluated on the use of an appropriate checklist.

Assessing proper checklist use depends upon the specific Task. In all cases, the evaluator should determine whether the applicant demonstrates CRM, appropriately divides attention, and uses proper visual scanning. In some situations, reading the actual checklist may be impractical or unsafe. In such cases, the evaluator should assess the applicant's performance of published or recommended immediate action "memory" items along with their review of the appropriate checklist once conditions permit.

In a single-pilot airplane, the applicant should demonstrate the crew resource management (CRM) principles described as single-pilot resource management (SRM). Proper use depends on the specific Task being evaluated. If the use of the checklist while accomplishing elements of an Objective would be either unsafe or impractical in a single-pilot operation, the applicant should review the checklist after accomplishing the elements.

Positive Exchange of Flight Controls

A clear understanding of who has control of the aircraft must exist. Prior to flight, the pilots involved should conduct a briefing that includes reviewing the procedures for exchanging flight controls.

The FAA recommends a positive three-step process for exchanging flight controls between pilots:

- When one pilot seeks to have the other pilot take control of the aircraft, they will say, "You have the flight controls."
- The second pilot acknowledges immediately by saying, "I have the flight controls."
- The first pilot again says, "You have the flight controls," and visually confirms the exchange.

Pilots should follow this procedure during any exchange of flight controls, including any occurrence during the practical test. The FAA also recommends that both pilots use a visual check to verify that the exchange has occurred. Doubt as to who is flying the aircraft should not occur.

Use of Distractions

Numerous studies indicate that many accidents have occurred when the pilot has been distracted during critical phases of flight. The evaluator should incorporate realistic distractions during the flight portion of the practical test to evaluate the pilot's situational awareness and ability to utilize proper control technique while dividing attention both inside and outside the flight deck.

Aeronautical Decision-Making, Risk Management, Crew Resource Management, and Single-Pilot Resource Management

Throughout the practical test, the evaluator must assess the applicant's ability to use sound aeronautical decision-making procedures in order to identify hazards and mitigate risk. The evaluator must accomplish this requirement by reference to the risk management elements of the given Task(s), and by developing scenarios that incorporate and combine Tasks appropriate to assessing the applicant's risk management in making safe aeronautical decisions. For example, the evaluator

may develop a scenario that incorporates weather decisions and performance planning.

In assessing the applicant's performance, the evaluator should take note of the applicant's use of CRM and, if appropriate, SRM. CRM/SRM is the set of competencies that includes situational awareness, communication skills, teamwork, task allocation, and decision-making within a comprehensive framework of standard operating procedures (SOP). SRM specifically refers to the management of all resources onboard the aircraft, as well as outside resources available to the single pilot.

If an applicant fails to use aeronautical decision-making (ADM), including SRM/CRM, as applicable in any Task, the evaluator will note that Task as failed. The evaluator will also include the ADM Skill element from the Flight Deck Management Task on the Notice of Disapproval of Application.

Multiengine Considerations

On multiengine practical tests, where the failure of the most critical engine after liftoff is required, the evaluator must consider local atmospheric conditions, terrain, and type of aircraft used. The evaluator must not simulate failure of an engine until attaining an altitude of at least 400 feet AGL and at least minimum single-engine speed (V_{SSE}), best single-engine angle-of-climb speed (V_{XSE}), or best single-engine rate-of-climb (V_{YSE}).

The applicant must supply an airplane that does not prohibit the demonstration of feathering the propeller in flight. However, an applicant holding an unrestricted AMEL rating may take a practical test for the addition of an AMES rating in an AMES without propeller feathering capability. Practical tests conducted in a flight simulation training device (FSTD) can only be accomplished as part of an approved curriculum or training program pursuant to 14 CFR part 61, section 61.64. Any limitations or powerplant failure will be noted in that program.

For safety reasons, when the practical test is conducted in an airplane, the applicant must perform Tasks that require feathering or shutdown only under conditions and at a position and altitude where it is possible to make a safe landing on an established airport if there is difficulty in unfeathering the propeller or restarting the engine. The evaluator must select an entry altitude that will allow the single-engine demonstration Tasks to be completed no lower than 3,000 feet AGL or the manufacturer's recommended altitude (whichever is higher). If it is not possible to unfeather the propeller or restart the engine while airborne, the applicant and the evaluator should treat the situation as an emergency. At altitudes lower than 3,000 feet AGL, engine failure should be simulated by reducing throttle to idle and then establishing zero thrust.

Engine failure (simulated) during takeoff should be accomplished prior to reaching 50 percent of the calculated V_{MC} .

Single-Engine Considerations

For safety reasons, the evaluator will not request a simulated powerplant failure in a single-engine airplane unless it is possible to safely complete a landing.

Appendix 3: Aircraft, Equipment, and Operational Requirements & Limitations

Aircraft Requirements & Limitations

If the multiengine airplane used for the practical test does not publish a V_{MC} , then the “Limited to Center Thrust” limitation will be added to the certificate issued from this check, unless the applicant has previously demonstrated competence in a multiengine airplane with a published V_{MC} .

If the aircraft has inoperative equipment and can be operated in accordance with 14 CFR part 91, section 91.213, it must be determined if any inoperative instruments or equipment are required to complete the practical test. The inoperative equipment must not interfere with practical test requirements.

Equipment Requirements & Limitations

The aircraft must meet the requirements as outlined in 14 CFR part 61, section 61.45.

To assist in management of the aircraft during the practical test, the applicant is expected to demonstrate automation management skills by utilizing installed, available, or airborne equipment such as autopilot, avionics and systems displays, and/or a flight management system (FMS). The evaluator is expected to test the applicant's knowledge of the systems that are available or installed and operative during both the ground and flight portions of the practical test. If the applicant has trained using a portable electronic flight bag (EFB) to display charts and data and wishes to use the EFB during the practical test, the applicant is expected to demonstrate appropriate knowledge, risk management, and skill appropriate to its use.

If the practical test involves maneuvering the aircraft solely by reference to instruments, the applicant is required by 14 CFR part 61, section 61.45(d)(2) to provide an appropriate view limiting device acceptable to the Administrator. The applicant and the evaluator should establish a procedure as to when and how this device should be donned and removed and brief this procedure before the flight. This device must prevent the applicant from having visual reference outside the aircraft, but it must not restrict the evaluator's ability to see and avoid other traffic. The use of the device does not apply to specific elements within a Task when there is a requirement for visual references.

Use of Flight Simulation Training Devices (FSTD)

Applicants for a pilot certificate or rating can accomplish all or part of a practical test or proficiency check in an FSTD qualified under 14 CFR part 60, which includes full flight simulators (FFS) or flight training devices (FTD), only when conducted within an FAA-approved training program. Each operational rule part identifies additional requirements for the approval and use of FSTDs in an FAA-approved training program.

Credit for Pilot Time in an FSTD

14 CFR part 61 and part 141 specify the minimum experience requirements for each certificate or rating sought. 14 CFR part 61 and the appendices to part 141 specify the maximum amount of FFS or FTD flight training time an applicant can apply toward those experience requirements.

Use of Aviation Training Devices (ATD)

Applicants for a pilot certificate or rating cannot use an ATD to accomplish a practical test, a 14 CFR part 61, section 61.58 proficiency check, or the flight portion of a 14 CFR part 61, section 61.57 flight review. An ATD is defined in 14 CFR part 61, section 61.1.

The FAA's General Aviation and Commercial Division evaluates and approves ATDs as permitted under 14 CFR part 61, section 61.4(c) and FAA Order 8900.1. Each ATD is then issued an FAA letter of authorization (LOA) that is valid for 60 calendar months. The LOA for each ATD lists the pilot time credit allowances and associated limitations.

The Pilot Training and Certification Group public website provides [a list of the FAA-approved ATDs](#) and the associated manufacturer.

Credit for Pilot Time in an ATD

14 CFR part 61 and part 141 specify the minimum experience requirements for each certificate or rating sought. 14 CFR part 61 and the appendices to part 141 specify the maximum amount of ATD flight training time an applicant can apply

toward those experience requirements. The LOA for each FAA-approved ATD lists the pilot time credit allowances and the associated limitations.

Evaluators must request an applicant to provide a copy of the manufacturer's LOA when using ATD flight training time credit to meet the minimum experience requirements for an airman pilot certificate, rating, or privilege.

Operational Requirements, Limitations, & Task Information

VII. Slow Flight and Stalls

Task A. Maneuvering During Slow Flight

Evaluation criteria for this Task should recognize that environmental factors (e.g., turbulence) may result in a momentary activation of stall warning indicators such as the stall horn. If the applicant recognizes the stall warning indication and promptly makes an appropriate correction, a momentary activation does not constitute unsatisfactory performance on this Task. As with other Tasks, unsatisfactory performance would arise from an applicant's continual deviation from the standard, lack of correction, and/or lack of recognition.

Task B. Power-Off Stalls

Evaluation criteria for a recovery from an approach to stall should not mandate a predetermined value for altitude loss and should not mandate maintaining altitude during recovery. Proper evaluation criteria should consider the multitude of external and internal variables that affect the recovery altitude.

Task C. Power-On Stalls

In some high-performance airplanes, the power setting may have to be reduced below the ACS guidelines power setting to prevent pitch attitudes greater than 30° nose up. Evaluation criteria for a recovery from an approach to stall does not mandate a predetermined value for altitude loss and does not mandate maintaining altitude during recovery. Proper evaluation criteria considers the multitude of external and internal variables that affect the recovery altitude.

IX. Emergency Operations

Task D. Emergency Equipment and Survival Gear

For airplanes that include a ballistic parachute, applicants must follow the manufacturer's procedures for arming and disarming the system before and after flight. Testing of an applicant's knowledge regarding how and when to use the system and how to manage associated risks may include simulation and briefing of procedures but not actual deployment of the system.

Task E. Engine Failure During Takeoff Before V_{MC} (Simulated) [Airplane, Multiengine Land (AMEL); Airplane Multiengine Sea (AMES)]

Engine failure (simulated) during takeoff should be accomplished prior to reaching 50 percent of the calculated V_{MC} .

Task F. Engine Failure After Liftoff (Simulated) (AMEL, AMES)

The evaluator must not simulate failure of an engine until attaining an altitude of at least 400 feet AGL and at least minimum single-engine speed (V_{SSE}), best single-engine angle-of-climb speed (V_{XSE}), or best single-engine rate-of-climb (V_{YSE}).

X. Multiengine Operations

Task B. V_{MC} Demonstration (AMEL, AMES)

Airplanes with normally aspirated engines will lose power as altitude increases because of the reduced density of the air entering the induction system of the engine. This loss of power will result in a V_{MC} lower than the stall speed at higher altitudes. Therefore, recovery should be made at the first indication of loss of directional control, stall warning, or buffet. Do not perform this maneuver by increasing the pitch attitude to a high angle with both engines operating and then reducing power on the critical engine. This technique is hazardous and may result in loss of airplane control.

Task C. One Engine Inoperative (Simulated) (solely by Reference to Instruments) During Straight-and-Level Flight and Turns (AMEL, AMES)

This Task is not required if an instrument-rated applicant has previously demonstrated instrument proficiency in a multiengine airplane, or if the applicant does not hold an Instrument Airplane Rating. If an applicant holds both a single- and multiengine rating on a pilot certificate, but has not demonstrated instrument proficiency in a multiengine aircraft, that airman's certificate must bear a limitation indicating that multiengine flight is permitted in visual flight rules (VFR) conditions only.

Task D. Instrument Approach and Landing with an Inoperative Engine (Simulated) (solely by Reference to Instruments) (AMEL, AMES)

This Task is not required if an instrument-rated applicant has previously demonstrated instrument proficiency in a multiengine airplane, or if the applicant does not hold an Instrument Airplane Rating. If an applicant holds both a single- and multiengine rating on a pilot certificate, but has not demonstrated instrument proficiency in a multiengine aircraft, that airman's certificate must bear a limitation indicating that multiengine flight is permitted in visual flight rules (VFR) conditions only.